Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Plasmonic sensor for on-site detection of chemical and microbiological pollutants in process water			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Water/Soil	Environmental Facility Technologies	Industrial Biotechnology	Bio-based Agricultur/Livestock/Aquatic Preparations
Concept of Proposed Technology	Increasing water scarcity, heightened public scrutiny of pollutant discharges, and progressively stricter regulatory requirements for the reduction of trace substances demand new concepts for detecting chemical and microbiological pollutants. Conventional monitoring approaches—periodic grab sampling and delayed laboratory analysis—cannot capture transient events or provide the temporal resolution needed for reliable process control and compliance. There is a pressing need for continuous, digital, and selective in situ detection that delivers real-time, actionable data across diverse and variable water matrices. To address these needs an innovative photonic sensing platform is developed that leverages localized surface plasmon resonance (LSPR) in engineered nanostructures to convert the specific binding of chemical or microbiological pollutants to surface-immobilized receptors into a quantifiable optical signal. The label-free system is modular, digital, and designed for continuous 24/7 operation under real-world process conditions, enabling comprehensive monitoring of application-tailored trace			

	substances in municipal and industrial (waste)water. Continuous online measurement generates real-time trends and event alerts, enabling rapid identification of transient spikes, process deviations, or emerging risks that conventional grab sampling would miss. The modular architecture of the sensor chip with multiple sensitive nanostructured areas supports application-specific target panels, allowing reconfiguration for application- or site-specific combinations of target analytes, matrices, and regulatory priorities. The platform interfaces seamlessly with plant control systems to enable demand-driven, resource-efficient operation, for example: (1) In municipal wastewater treatment, the system enables continuous effluent monitoring and adaptive control of advanced treatment processes such as ozonation or activated carbon, improving removal efficiency while optimizing energy and consumables. Real-time insights support compliance with tightening discharge limits, enhance operational resilience, and provide early warning of treatment upsets or contamination events. The platform can also be deployed in upstream sewers or at critical nodes to detect illicit discharges and protect downstream assets. (2) In industrial settings, the technology enhances process water and discharge monitoring across sectors with complex, variable matrices. It supports water reuse and closed-loop strategies by verifying quality in real time, reducing reliance on batch analyses, and enabling rapid interventions when thresholds are exceeded. The system can be tailored to sector-specific pollutants, facilitating targeted monitoring for pharmaceuticals, chemicals, pesticides, herbicides, microbiological parameters like E.coli, Legionella or pathogens relevant in green houses or aquaculture. (3) Deployed as a network of sensors, the system offers spatially resolved monitoring to inform watershed management, risk assessments, and evidence-based policy making.
Rationale for Joint R&D Project	The plasmonic sensor technology, especially the fabrication of the sensor substrates by nano-imprint lithography as well as the biofunctionalization of the sensor substrates with the application-secific recognition elements, is strongly related to IKTS core competencies. As water supplies in Korea, as in Europe, are coming under increasing pressure, public awareness of pollutant emissions into waterways is growing, and requirements for reducing trace substances are becoming ever stricter, there is a great need for 24/7 monitoring of application-relevant trace substances. Joint R&D is required because the successful development and commercialization of the technology depend on the combination of both partners' strengths: the research institute contributes advanced know-how in sensor substrate fabrication, functionalization and sensor system concepts, while Korean SMEs provide access to relevant industrial/municipal use cases, know-how on synthesis of application-specific recognition elements, water treatment lines for pilot implementation, and the market knowledge necessary for adapting and scaling the solution to the Korean and wider Asian markets. Only through close collaboration can the technology be validated under real Korean industrial conditions and transferred into commercially viable products and services.
Key Words	1. surface plasmon resonance spectroscopy 2. biochemical assay/recognition element 3. pollutants of emerging concern 4.on-site/online monitoring 5.one-health approach

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Sensor and assay development: - Selection of pollutants/microbiological parameters relevant for a specific Korean pilot site. Search for commercially available recognition elements or synthesis thereof (e.g. aptamers, antibodies, molecular imprinted polymers) - Development of biochemical assays for immobilization and regeneration of the recognition elements on the gold sensor surface Lab-Scale Test: - Quantitative determination of pollutant concentrations in water samples from the pilot site, before and after a water treatment step; Measurements in the laboratory in comparison to state-of-the-art standard technologies like liquid chromatography One-Site test campaign - Integration of the plasmonic measurement system in the water treatment process at the pilot site and run of a test campaign - Development of a concept for regulating water treatment using measurement data from the sensor unit - Evaluation of energy and operational savings (efficiency gains) and environmental sustainability			
R&D Target Specifications	- Multiplex (parallel) detection of up to 3 pollutants in the application-specific relevant concentration ranges, i.e. Diclofenac as an example of a pharmaceutical indicator substance can be detected in the range of 1-10 µg/L - Measurement duration: the measurement of the 3 pollutants' concentration takes 15-30 min - Measurement location: the concentration measurement takes place automatically directly at the inlet/outlet of a water treatment system. - A measurement campaign lasting at least 1 month will be carried out - Specific costs for energy and operating cost savings were determined			
R&D Plan	The R&D plan is structured into consecutive phases, combining test and development of recognition elements and biochemical assays, testing of the measurement unit in lab-scale, measurement at the pilot site and evaluation of efficiency gains/environmental sustainability Pre-study (month 0-6) - Definition of 5-7 substances relevant at the pilot site - Determination of existing contamination levels with the target substances by reference methods like high-pressure liquid chromatography - Literature and market research on (commercially) available recognition elements for the specified target substances - Selection of 3 target analytes, for which there is a commercial recognition element for at least 2 - Pre-test of one chosen commercial recognition element - Methodology: literature review, benchmarking of existing contamination levels, preliminary lab tests - Milestone: Definition of 3 target analytes and their application-relevant concentration ranges Development of biochemical recognition elements and assays (month 7 – 15) - Development/Synthesis of a recognition element (aptamer/antibody/molecular imprinted polymer) for one analyte for which no recognition element is commercially available, including tests on affinity and specificity - Development of biochemical protocols for immobilization of the recognition elements (aptamer/antibody/MIP – commercially available and later on the synthesized new			

	recognition element) on nanostructured gold surfaces - Evaluation of the binding of the target analytes from spiked samples to the functionalized sensor surfaces - Development of assay conditions for regeneration of the sensor surface to allow for repetitive use - Milestone: Binding and regeneration of at least 1 target analyte is approved on the plasmonic sensor surface Lab-scale testing (month 12 – 20) - Optimization of functionalization and assay parameters with regard to sensitivity and measuring range by varying, for example, incubation time, concentrations, volume flows - Determination of calibration curves - Milestone: calibration curves for at least 2 target analytes are determined meeting the applicant-relevant concentration ranges Integration of measurement unit at pilot site (month 15 – 23) - Requirements analysis for the integration of the plasmonic sensor unit at the pilot site (interface to the water treatment system, environmental conditions) - Derivation of a concept for the automation of sampling - Preliminary tests at the water treatment system without the plasmonic sensor unit for recording the actual status (Methodology: lab routine established at a local reference lab, liquid chromatography) - Development of a concept for regulating the water treatment - Integration of the plasmonic sensor unit at the pilot site - Milestone: plasmonic sensor unit is integrated at pilot site and ready for test campaign Test campaign at pilot site (month 24 – 27) - Operation of the water treatment system with on-site monitoring by the plasmonic sensor unit, including sampling and laboratory analysis (in the pilot sites' reference laboratory). - Use of the plasmonic sensor unit for continuous monitoring of the water treatment; optimization of measurement operations; - Use of the developed control concept to reduce energy and operation costs - Milestone: successful demonstration of the on-site monitoring in a relevant industrial environment at a Korean pilot site Evaluation of smart water treatment (month 27 – 30) - Comparison of water treatment efficiency and pollutant load with/without on-site monitoring - Presentation and quantification of pilot plant-related savings in personnel, energy (including indirect GHG emissions), and operating resources - Collect data in database for correlation analysis using AI - Quantitative assessment of the techno-economic and ecological potential within the reference sector and in cross-sector application areas - Methodology: implementation of data acquisition and AI-based analysis tools, correlation of water treatment process parameters with pollutant load. - Milestone: proof-of-concept demonstration of data-driven process optimization

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Search in literature and market for recognition elements - Development, optimization and test of biochemical assays - Functionalization of plasmonic sensor substrates with developed protocols - Calibration of sensor substrates with spiked samples - Manufacture and provision of plasmonic sensor substrates for lab tests and test campaign - Support for integration of the sensor unit at the pilot site - Support during test campaign
Expected R&R of Korean SME Partner	- Development/synthesis of recognition element (aptamer/ antibody/ molecular imprinted polymer) for specific binding of a small pollutant molecule or bacteria on gold sensor substrate; provision of the recognition element for functionalization on sensor substrates - Provision of a pilot site and coordination of the test campaign there - Integration and installation of the plasmonic sensor unit at the pilot site - Development of a concept for regulating the water treatment at the pilot site - Conducting the test campaign, including coordination of reference measurements - Quantification of pilot plant-related savings

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological achievements: - Validated plasmonic sensing platform at TRL 7 for on-site monitoring of application-relevant trace substances in municipal and industrial (waste)water. - Modular sensor chip designs with multiple sensitive nanostructured areas and exchangeable target panels, enabling rapid adaptation to site- or sector-specific analyte sets - System-level interfaces and control concepts that link real-time sensing with plant operations for demand-driven, resource-efficient treatment (e.g., adaptive ozonation/activated carbon/disinfection) Commercialization potential: - Clear value proposition for utilities and industry: continuous compliance assurance, early warning of transient events, and optimized advanced treatment that lowers energy and consumables while safeguarding effluent quality. - Go-to-market readiness supported by: • A pilot-proven demonstrator operating in a relevant industrial environment in Korea. • Application-specific target panels for priority pollutants and microbiological parameters (e.g., pharmaceuticals, pesticides/herbicides, E. coli, Legionella, pathogens relevant to greenhouses or aquaculture). • Integration with plant control systems and secure data transmission to on-premise/cloud platforms. - Business model options combining hardware (sensor unit), consumables (functionalized sensor substrates), and digital services (data analytics, alarms, reporting), creating recurring revenue streams - Potential contributions to future standards for online micropollutant monitoring Broader industrial and economic impact: - Reduced risk of non-compliance and associated penalties via continuous effluent quality verification and early detection of process upsets or illicit discharges. - Enablement of water reuse and closed-loop strategies by providing real-time quality assurance, supporting resilience in the face of water scarcity. - Transferability across sectors (municipal utilities, pharmaceuticals/chemicals, food and beverage, microelectronics, agriculture/aquaculture), fostering cross-sector adoption and market growth. - Contribution to data-centric water management: continuous, traceable datasets support evidence-based decision-making, performance benchmarking, and lifecycle cost optimization. Environmental and societal impact: - Improved protection of receiving waters through continuous reduction of trace substance discharges and timely response to contamination events. - Support for One Health objectives by addressing chemical and microbiological risks at the water–environment–health interface. - Foundation for watershed-scale monitoring via sensor networks, enabling spatially resolved management, risk assessments, and policy development

6. References

(1) Steinke et al. Sensors & Actuators: B. Chemical 288 (2019) 594–600. https://doi.org/10.1016/j.snb.2019.02.069